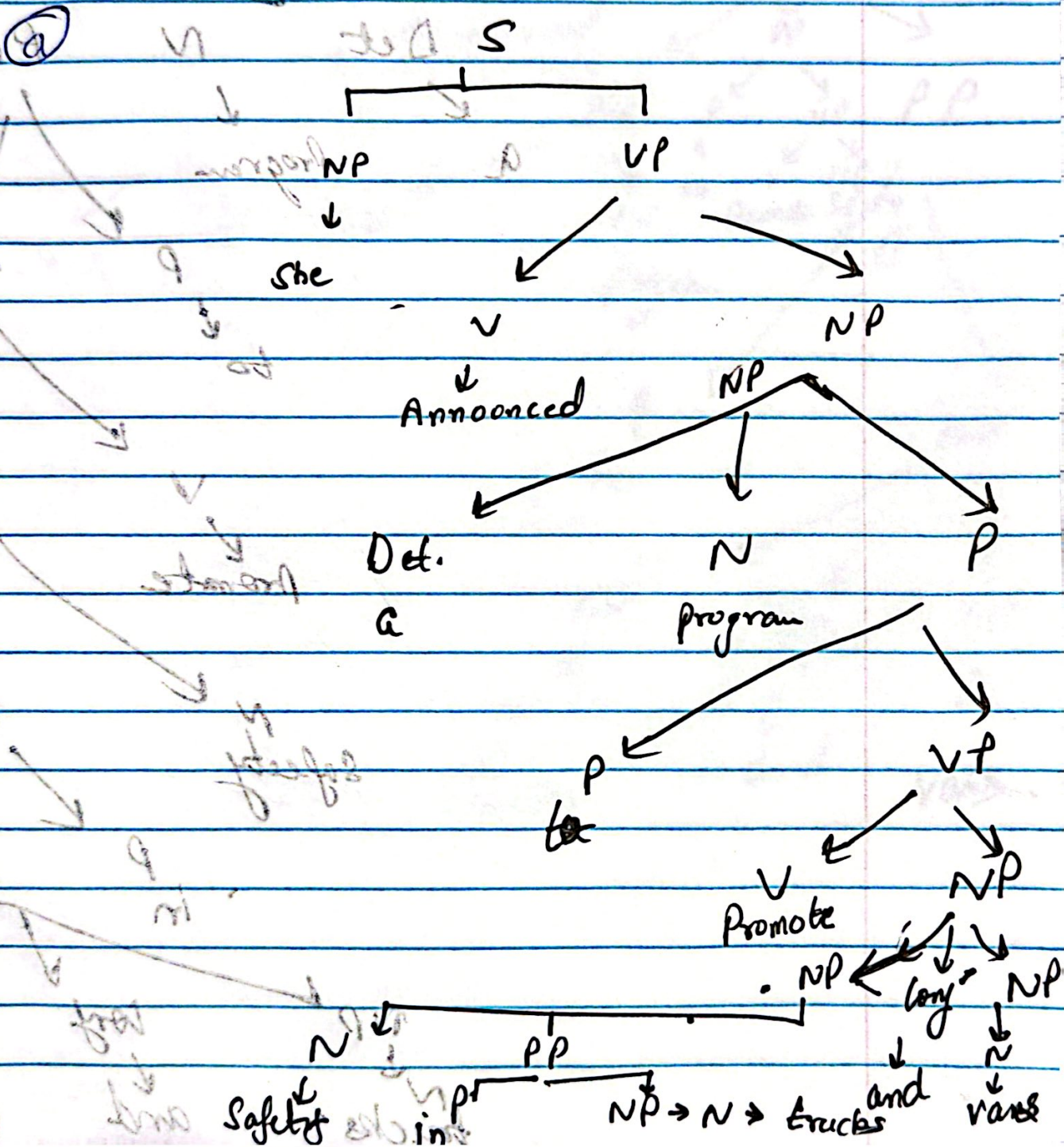


Assignment 2

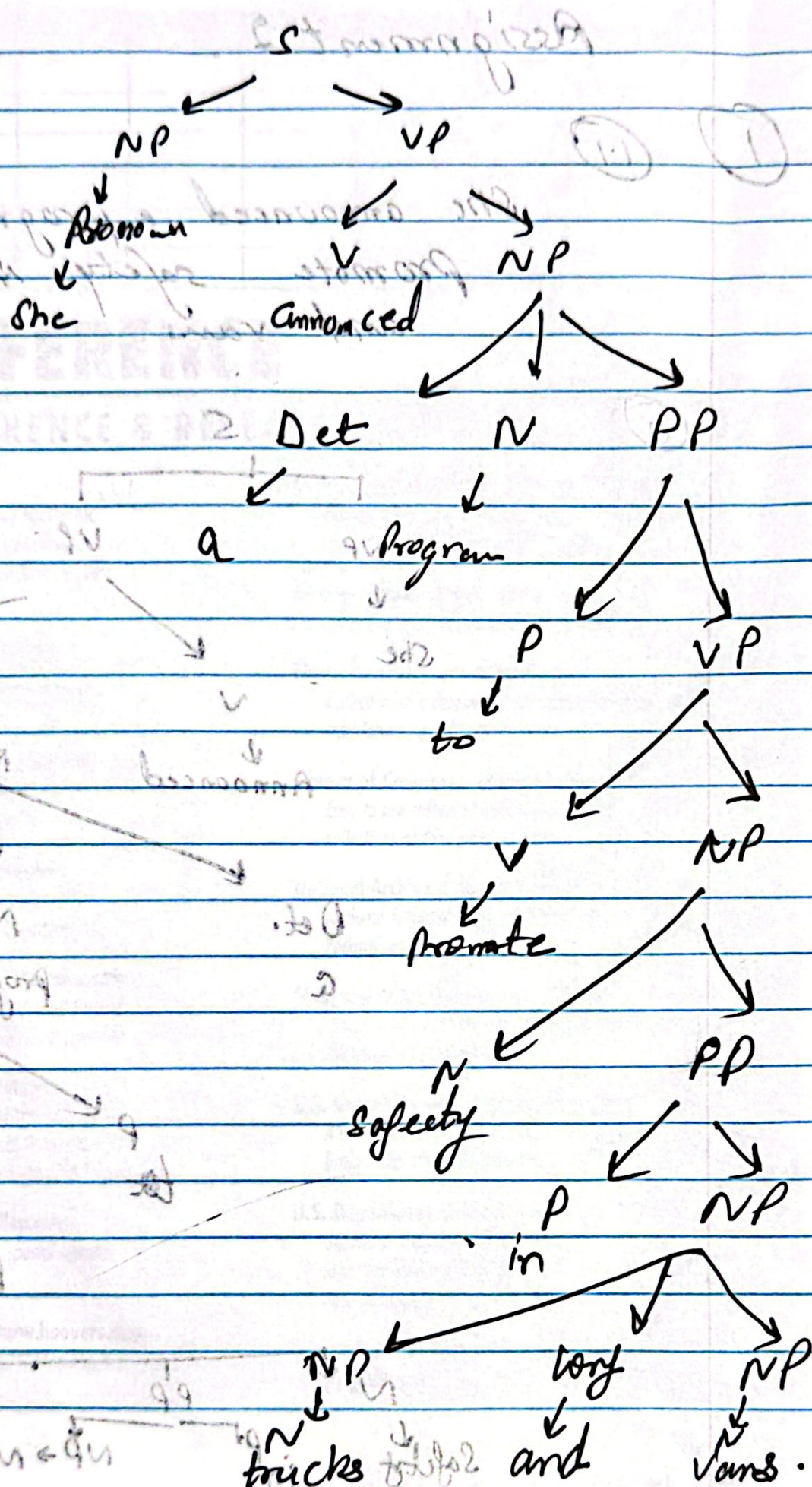
①

①.①

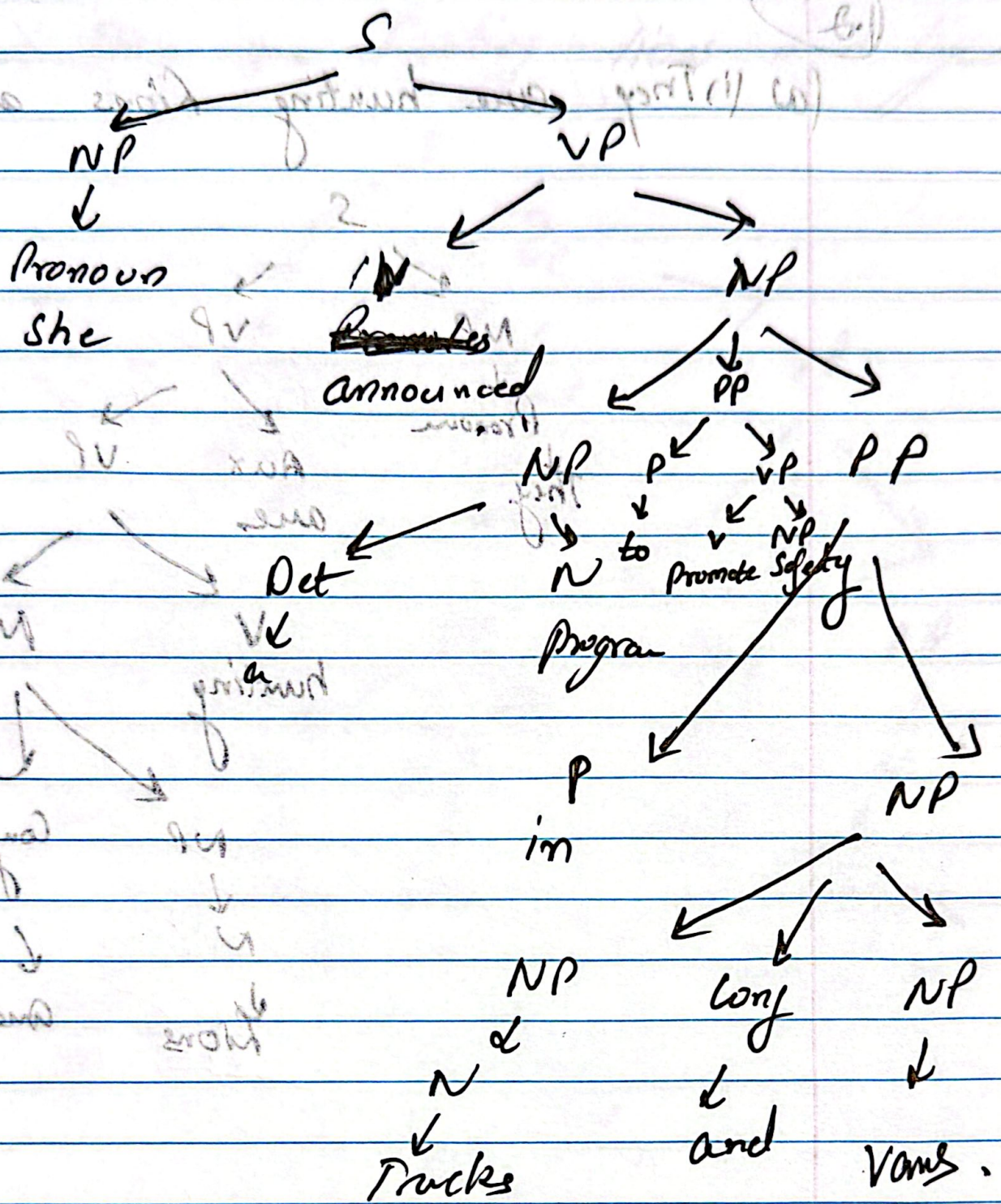
She announced a program to promote safety in trucks and vans.



⑥

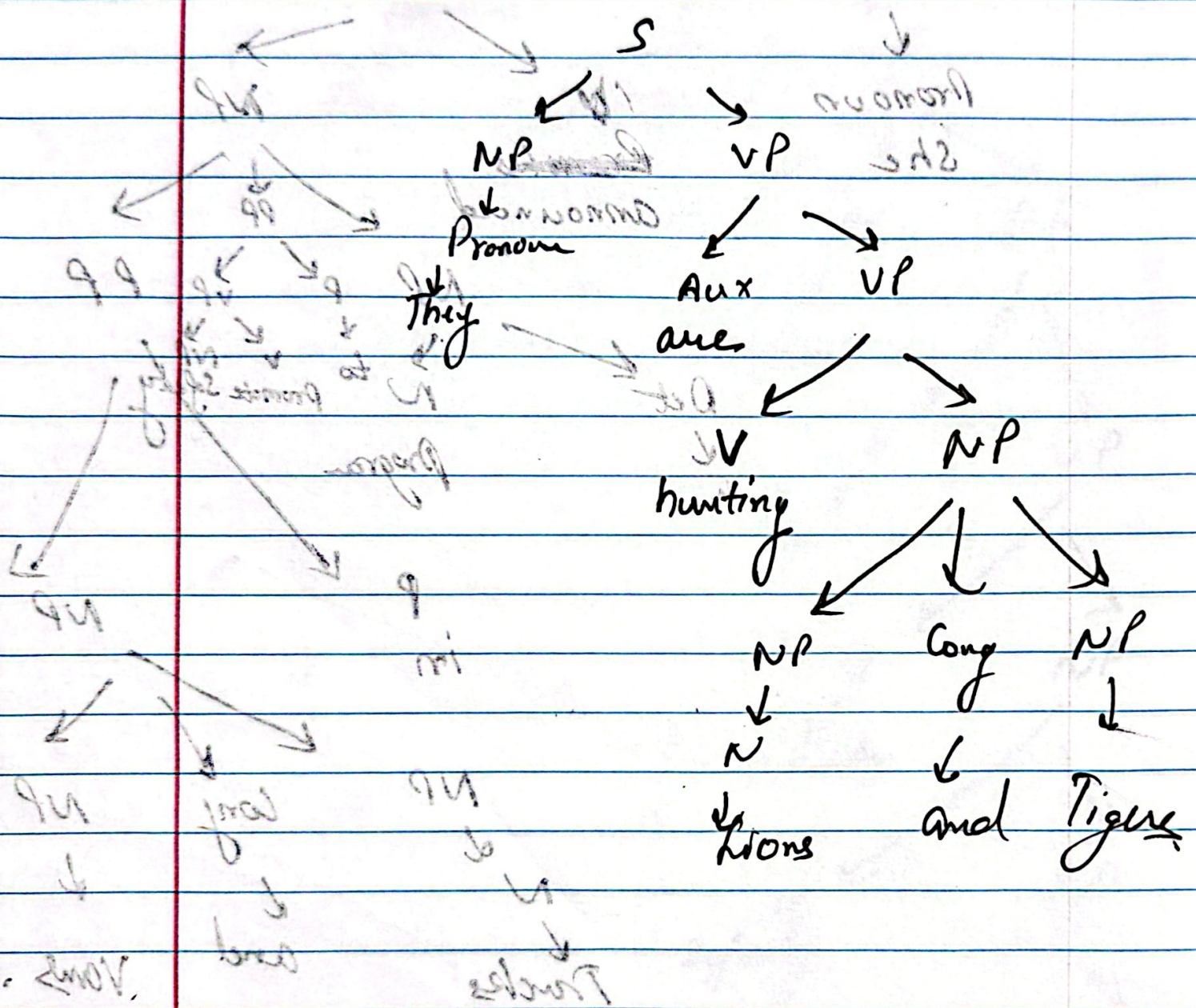


(C)

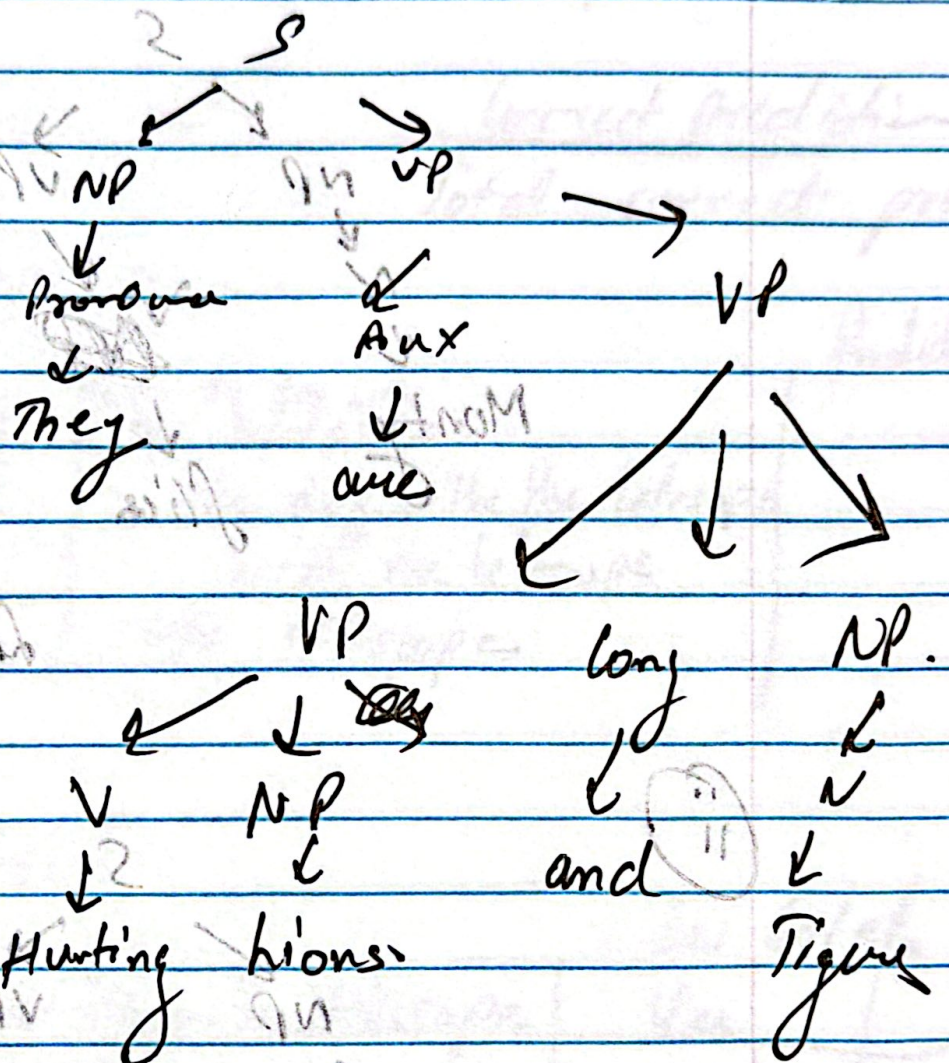


1.2

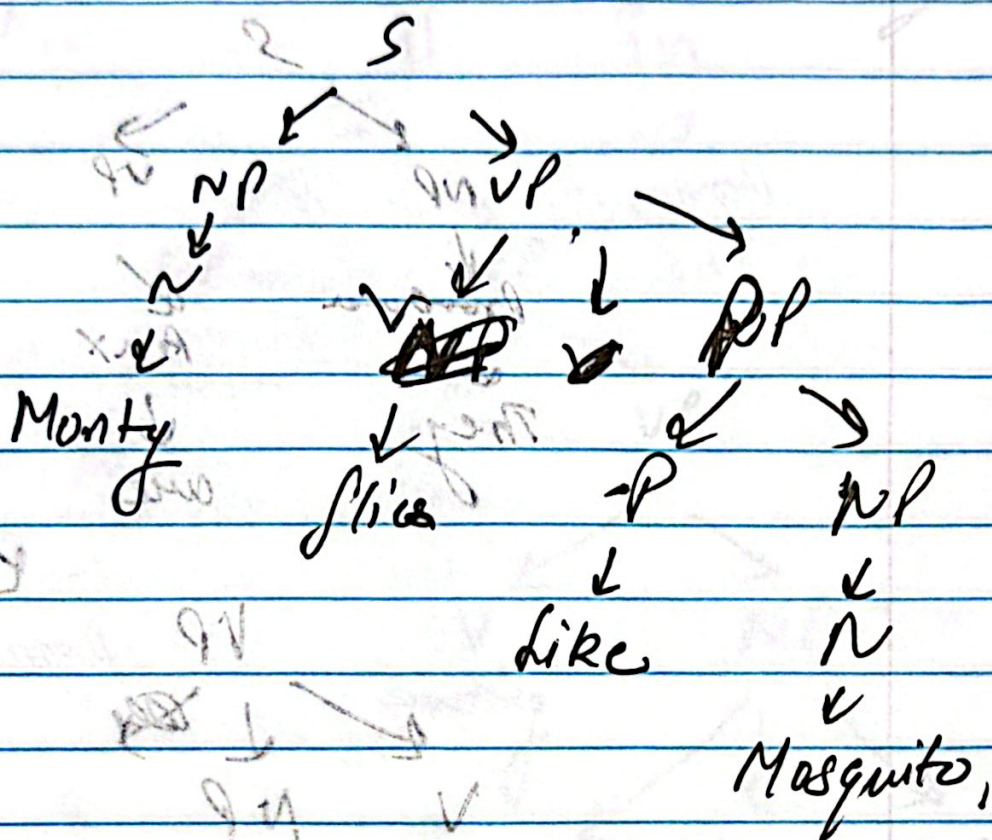
(a) (i) They are hunting lions and tigers



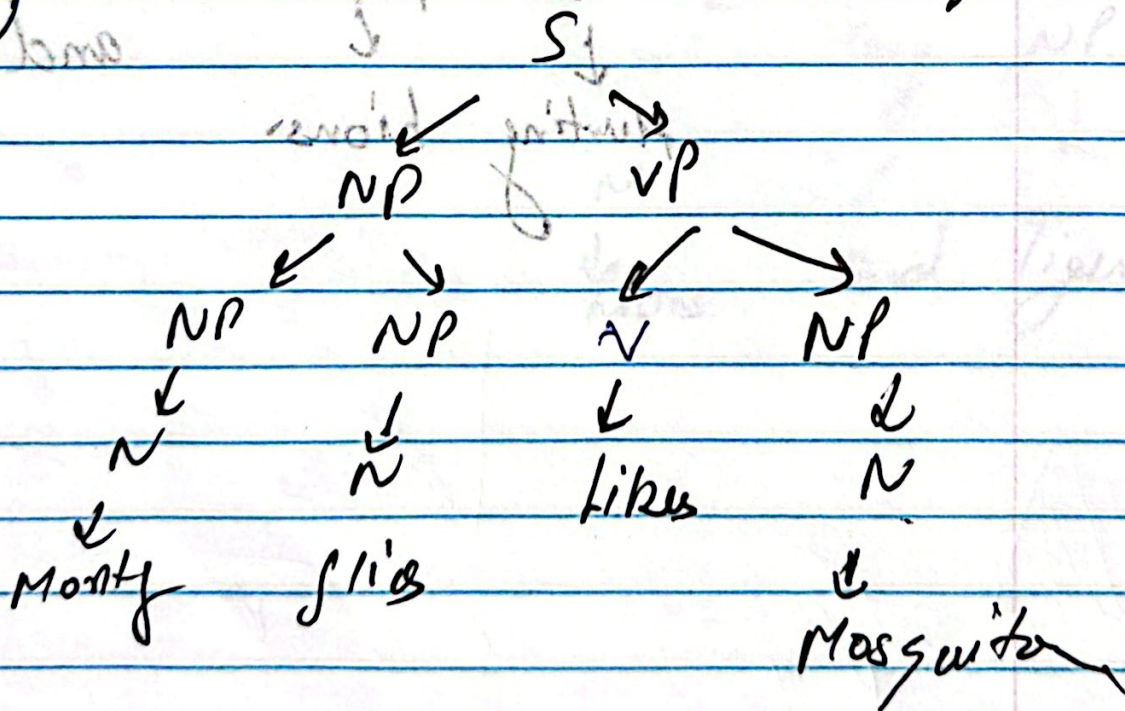
(a) (ii) They are hunting lions and tigers



(b) (i) Monty slices like mosquitoes (ii) (a)



(ii)



②

$$\text{Precision} = \frac{\text{correct predictions}}{\text{Total prediction}}$$

$$\text{Recall} \Rightarrow \frac{\text{correct prediction}}{\text{Total correct pred}}$$

Gold Tree

			Predicted
1	VP	Saw the dog with the telescope	
2	NP	the dog with the telescope	
3	PP	with the telescope	
4	NP	the telescope	

Predicted Tree

			IN Gold
1	VP	Saw the dog with telescope	Yes
2	NP	the dog with the telescope	Yes
3	NP	the dog	No
4	PP	with the telescope	Yes
5	NP	the telescope	Yes

$$\text{Precision} = \frac{4}{5} = 80\%$$

$$\text{Recall} \Rightarrow \frac{4}{4} = 100\%$$

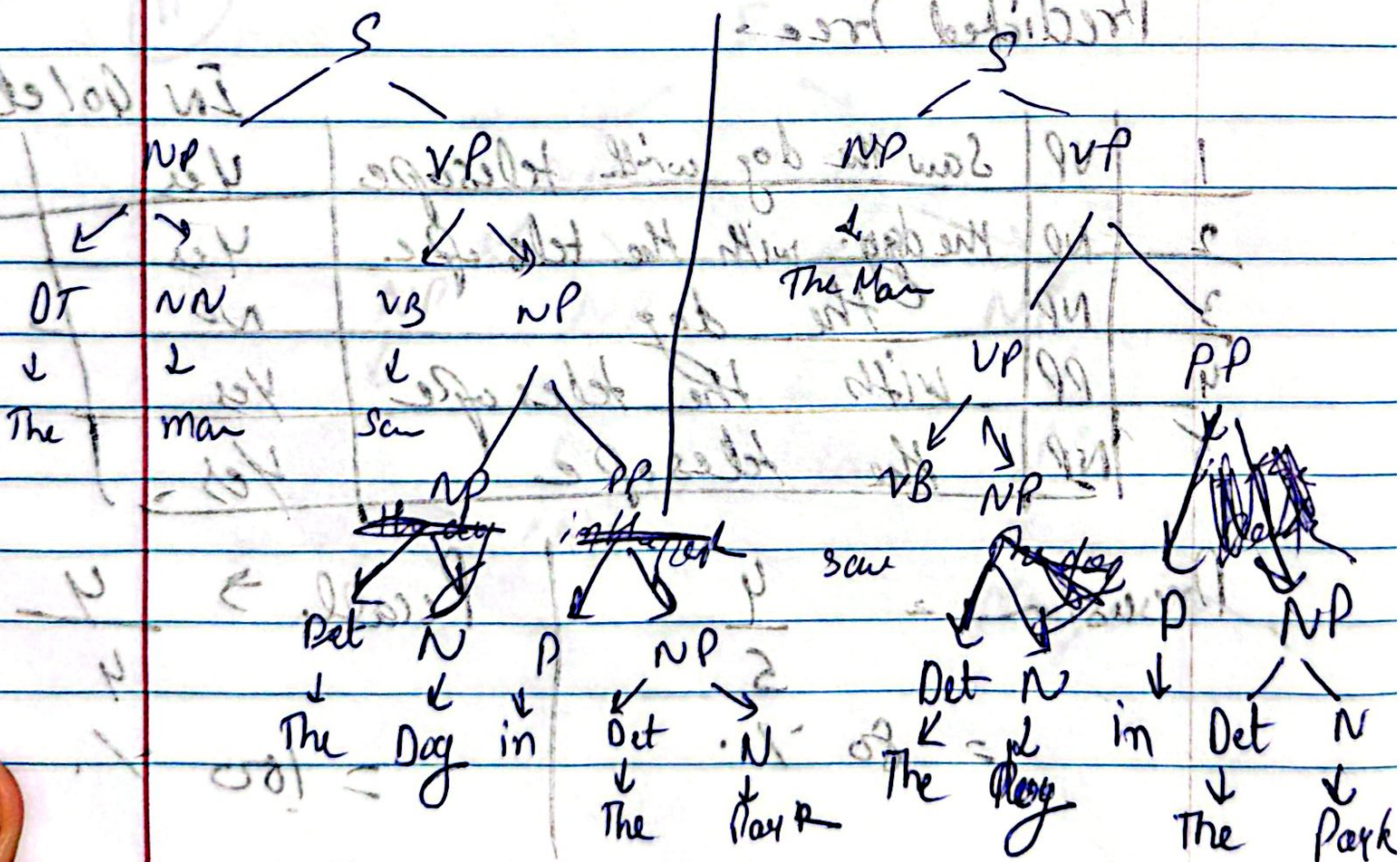
③ ③.1 Two parse trees can be made

① The dog in the park, i.e.,

the dog was in the park

② saw the dog, in the park

the seeing man was in the park



3.2 There can be multiple (Five) parse trees.

five 5 Possible Interpretations:

① Both PP's to the verb phrase
[The man] [saw the dog] [in the park] [with the cat]

→ Man, in the park and with the cat saw the dog.

② [The man] [saw [the dog in the park] [with the cat]]
→ dog in the park as one group.

③ [The man] [saw the dog] [in [the park with cat]]

→ Park has the cat, and the man saw the dog.

④ [The man] [saw [the dog in the park] [with the cat]]
→ The man saw the dog which was in the park with cat. (Dog was with cat)

⑤ [The man] [saw [the dog in the park] with the cat]
→ The man saw the dog which was in the park, and the park also had the cat.

Reasoning: Each PP after the verb can attach to the VP or to any noun to its left, but attachments cannot cross. Counting by the top rule of the VP (either $VP \rightarrow VB NP$ with all PPs inside the NP, or $VP \rightarrow VP PP$ splitting off a final PP chunk) gives the Catalan recurrence, so n PPs give $C(n+1)$ trees.

(3.3) with k PPs, the number of parse trees is the $(k+1)$ Catalan number.

$$so \quad C_k = \frac{2k!}{(k+1)! (k)!}$$

So Answer

$$would be = C_{k+1} = \frac{[2(k+1)]!}{(k+2)! (k+1)!}$$

And this also gives answer for or verifies 3.1 & 3.2.
i.e

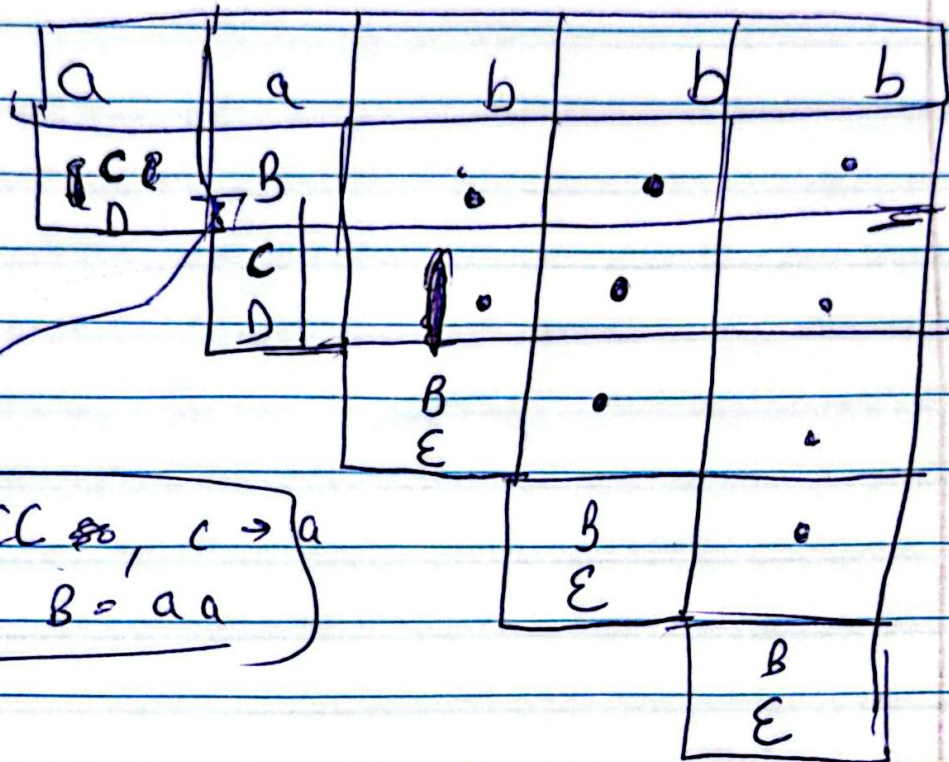
$$(3.1) \quad k=1, \quad so \quad \frac{(2(2))!}{3! \cdot 2!} = \frac{4!}{3! \cdot 2!} = \frac{4 \times 3!}{3! \cdot 2!} = 2$$

$$(3.2) \quad k=2, \quad so \quad \frac{(2(3))!}{4! \cdot 3!} = \frac{6!}{4! \cdot 3!} = \frac{6 \times 5 \times 4!}{3 \times 2 \times 1 \times 4!} = 5$$

so this verifies the formula.

$S \rightarrow AB$, $A \rightarrow BC$, $B \rightarrow CC$, $C \rightarrow AB$, $D \rightarrow a$,
 $E \rightarrow b$, $B \rightarrow b$, $C \rightarrow a$.

Q



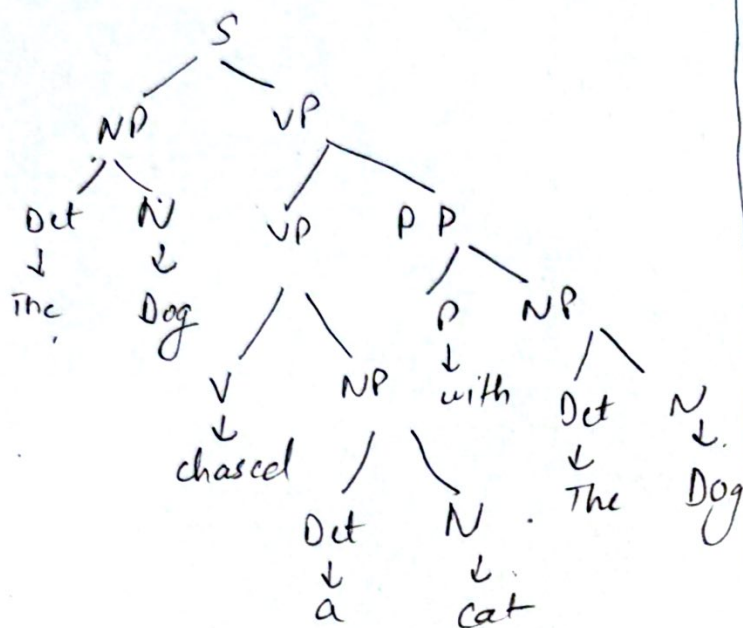
$B = CC$, $C \rightarrow a$
 So $B = aa$

No this, aabbba can't be formed from given grammar.

(5) (i) The dog chased a cat with the dog.

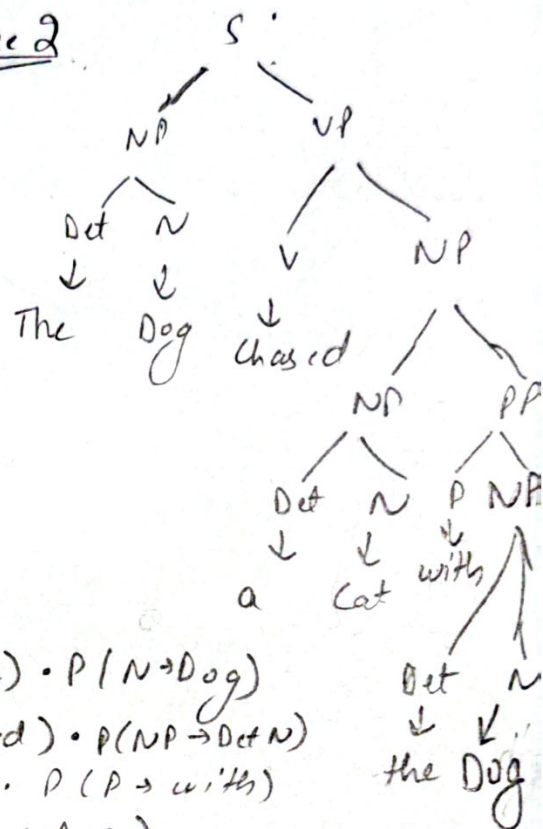
(ii) 1st way The dog chased a cat, with the dog.

Tree 1



(iii) The dog chased, a cat with the dog.

Tree 2



(B)

$$P(\text{Tree 1}) = P(S \rightarrow NP VP) \cdot P(NP \rightarrow Det N) \cdot P(Det \rightarrow The) \cdot P(N \rightarrow Dog) \\ \cdot P(VP \rightarrow VP PP) \cdot P(VP \rightarrow V NP) \cdot P(V \rightarrow chased) \cdot P(NP \rightarrow Det N) \\ \cdot P(Det \rightarrow a) \cdot P(N \rightarrow cat) \cdot P(PP \rightarrow P NP) \cdot P(P \rightarrow with) \\ \cdot P(NP \rightarrow Det N) \cdot P(Det \rightarrow The) \cdot P(N \rightarrow Dog).$$

$$= 1 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 1 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 1 \cdot 1 \cdot (0.5) \cdot (0.5)$$

$$= \left(\frac{1}{2}\right)^{11} \text{ or } (0.5)^{11} \text{ or } = 4.88 \times 10^{-4}$$

11 by for $P(\text{Tree 2}) = P(S \rightarrow NP VP) \cdot P(NP \rightarrow Det N) \cdot P(Det \rightarrow The) \cdot P(N \rightarrow Dog) \cdot P(VP \rightarrow VP PP) \cdot P(VP \rightarrow V NP) \cdot P(V \rightarrow chased) \cdot P(NP \rightarrow NP PP) \cdot P(NP \rightarrow Det N) \cdot P(Det \rightarrow a) \cdot P(N \rightarrow cat) \cdot P(PP \rightarrow P NP) \cdot P(P \rightarrow with) \cdot P(NP \rightarrow Det N) \cdot P(Det \rightarrow the) \cdot P(N \rightarrow cat)$

$$= \left(\frac{1}{2}\right)^{11} = (0.5)^{11} \text{ or } = 4.88 \times 10^{-4}$$

(C) Since $P(\text{Tree 1}) = P(\text{Tree 2})$, both are equally likely.

⑥ Sentence: I shot an elephant in my pajamas.

#	stack	Buffer	Action	Acc.
	[root]	[I, shot, an, eleph, in, my, pajamas]	—	—
①	[root, I]	[shot, an, elephant, in, my, pajamas, .]	Shift	shot → I
②	[root]	[shot, an, elephant, in my pajamas.]	Left arc	shot → I
③	[root, shot]	[an, elephant, in, my pajamas.]	Shift	
④	[root, shot, an]	[elephant, in, my, pajamas.]	Shift	
⑤	[root, shot]	[elephant, in, my, pajamas.]	Left arc	elephant → an (det)
6	[root, shot, elephant]	[in, my, pajamas.]	Shift	
7	[root, shot, elephant, in]	[my, pajamas.]	Shift	
8	[root, shot, elephant, in, my]	[pajamas.]	Shift	
9	[root, shot, elephant, in]	[pajamas.]	Left arc	pajamas → my (pos)
⑩	[root, shot, elephant]	[in.]	Right arc	in → pajamas (obj)
⑪	[root, shot]	[elephant, .]	Right arc	elephant → in (N mod)
⑫	[root]	[shot, .]	"	shot → elephant obj.
13	[root, shot]	[.]	Shift	Root → shot
14	[root]	[shot]	Right arc	shot → . (Punc)
⑬	[]	[root]	Right arc	Root → shot

